Lecture 35

Forced oscillations and Fourier coefficients

Uses.

Summary 2: Standing fact E supplies trigonometric addition and product identities, with angles in radians.
Summary 8: Scalar product, chain and trigonometric derivative rules hold.
Summary 16: Complex exponentials differentiate with real time; $\mathrm{Re}(Ce^{i\theta})$ has amplitude $|C|$ and phase $\arg C$ for $C \neq 0$.
Summary 26: Continuous integrands have definite integrals evaluated by primitives; substitution, parts and finite linearity hold.
Summary 29: A series converges through convergence of its partial sums.
Summary 30: A uniform limit of continuous functions is continuous.
Summary 34: Characteristic roots give the complete homogeneous family. Exponential forcing gives a particular solution, and every other solution differs by a homogeneous one. Positive damping makes that difference tend to zero.

Objectives. *A damped oscillator responds differently to different driving frequencies. We compute its amplitude and phase and locate the largest amplitude. Fourier coefficients then describe the frequencies of a periodic function. Orthogonality justifies the coefficient formulas, while a cited convergence theorem states what their infinite series represents. Finite sums of driving frequencies can be treated directly by linearity.*

By the end of this lecture you will be able to:

- Derive the amplitude and phase of the steady sinusoidal response for positive damping, including zero driving frequency.
- Locate the amplitude maximum and distinguish it from undamped resonant growth.
- Prove trigonometric orthogonality, compute real and complex Fourier coefficients, and apply the cited jump-value theorem under its hypotheses.
- Construct the unique periodic response to a real trigonometric polynomial with positive damping.

35.1 Sinusoidal forcing

Consider the normalised displacement equation

$$y'' + 2\gamma y' + \omega_0^2 y = F\cos(\omega t), \qquad \omega_0 > 0, \quad \gamma > 0, \quad F > 0, \quad \omega \geq 0.$$

The frequency ω_0 is the undamped angular frequency, and ω is the prescribed driving angular frequency. The coefficient γ measures damping as in Lecture 34. Write the characteristic polynomial as $P(\lambda) = \lambda^2 + 2\gamma\lambda + \omega_0^2$. The real and imaginary parts of its value at the driving exponent are

$$A = \omega_0^2 - \omega^2, \qquad B = 2\gamma\omega, \qquad P(\mathrm{i}\omega) = A + \mathrm{i}B.$$

This value never vanishes: at $\omega = 0$ one has $A = \omega_0^2 > 0$, and at $\omega > 0$ one has $B > 0$. The exponential forcing rule therefore gives the complex particular solution $Fe^{\mathrm{i}\omega t}/P(\mathrm{i}\omega)$. Real coefficients allow real parts to commute with the differential expression, so its real part solves the real equation. Complex division and multiplication give this real solution explicitly:

$$y_{\mathrm{st}}(t) = \frac{F(A\cos\omega t + B\sin\omega t)}{A^2 + B^2}. \tag{35.1}$$

The subscript denotes the *steady response*: every other solution differs from it by a homogeneous motion which tends to zero for future time. This last assertion follows from the positive-damping root cases of Lecture 34.

Define the amplitude R and the phase lag δ by

$$R = \frac{F}{\sqrt{A^2 + B^2}}, \qquad \cos\delta = \frac{A}{\sqrt{A^2 + B^2}}, \quad \sin\delta = \frac{B}{\sqrt{A^2 + B^2}}, \quad 0 \le \delta \le \pi.$$

The selected angle exists uniquely on that range because $B \ge 0$. The cosine subtraction formula makes Eq. (35.1) equal to $R\cos(\omega t - \delta)$. For $\omega = 0$, the convention gives $\delta = 0$ and the constant response F/ω_0^2. At $\omega = \omega_0$, the lag is $\pi/2$. The angle cannot in general be recovered from an unqualified $\arctan(B/A)$: the sign of A determines the quadrant, and A may vanish.

Example 35.1 (One forcing frequency)**.** Take $\omega_0 = 3$, $\gamma = 1$, $F = 2$ and $\omega = 2$. Here $A = 5$ and $B = 4$, so

$$y_{\mathrm{st}}(t) = \frac{10\cos 2t + 8\sin 2t}{41}, \qquad R = \frac{2}{\sqrt{41}}, \quad \delta = \arctan\frac{4}{5}.$$

The last expression has the correct quadrant because both A and B are positive. The homogeneous equation has roots $-1 \pm 2\sqrt{2}\,\mathrm{i}$. Thus every real solution is this steady response plus $e^{-t}(C\cos(2\sqrt{2}t) + D\sin(2\sqrt{2}t))$, for real C, D. The extra part tends to zero; its two coefficients are set by the initial data.

35.2 The amplitude maximum and resonance

The *amplitude response curve* is R as a function of $\omega \ge 0$, with F, γ and ω_0 fixed. Maximising R is equivalent to minimising its positive squared denominator. Set $s = \omega^2 \ge 0$ and complete the square:

$$(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2 = \bigl(s - (\omega_0^2 - 2\gamma^2)\bigr)^2 + 4\gamma^2(\omega_0^2 - \gamma^2).$$

If $\gamma < \omega_0/\sqrt{2}$, the square vanishes at the admissible positive value $s = \omega_0^2 - 2\gamma^2$. If $\gamma \ge \omega_0/\sqrt{2}$, its centre is nonpositive and the unique minimum on $s \ge 0$ is at zero. Consequently the frequency and amplitude of the maximum are

damping	maximising frequency	maximum amplitude
$0 < \gamma < \omega_0/\sqrt{2}$	$\sqrt{\omega_0^2 - 2\gamma^2}$	$F/(2\gamma\sqrt{\omega_0^2 - \gamma^2})$
$\gamma \ge \omega_0/\sqrt{2}$	0	F/ω_0^2.

The equality case belongs to the second row. A positive-frequency amplitude peak is called *displacement resonance* in this damped response curve. Its frequency is below ω_0. For the numerical example above the maximum occurs at $\sqrt{7}$, with amplitude $1/(2\sqrt{2})$, rather than at the example's driving frequency 2.

Figure 35.1 compares amplitude response curves with the same F and ω_0 but different positive damping.

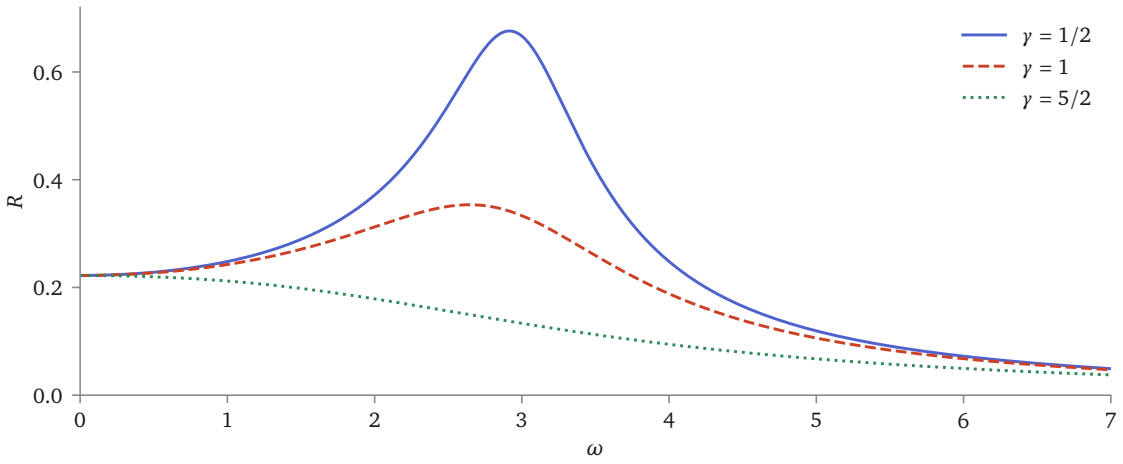


Figure 35.1: Amplitude response for $F = 2$ and $\omega_0 = 3$. The damping values $\gamma = 1/2$ and 1 give positive-frequency peaks; $\gamma = 5/2$ gives its maximum at zero.

The undamped equation at its natural frequency has a different behaviour. If $\gamma = 0$ and $\omega = \omega_0$, then $P(i\omega_0) = 0$. The simple-root forcing rule gives the real particular solution

$$y(t) = \frac{F}{2\omega_0} t \sin(\omega_0 t).$$

Its value and derivative at zero are both zero, so it is the response from rest. Differentiation gives $y'' + \omega_0^2 y = F\cos(\omega_0 t)$. The coefficient multiplying the sine grows with time; this motion has no bounded steady sinusoidal response. The positive-damping denominator formula was not asserted at this excluded parameter.

35.3 Orthogonality and Fourier coefficients

A periodic driving force need not be a single sine or cosine. To study a force represented by a *finite* sum of such waves, linearity lets us add the separately computed responses, one frequency at a time. Fourier coefficients tell us which sine and cosine terms occur in a chosen finite approximation. We first derive those coefficients; no interchange of an infinite sum with the differential equation is needed for that finite task. A function $f : \mathbb{R} \to \mathbb{R}$ is 2π*-periodic* if $f(x + 2\pi) = f(x)$ for every real x. We allow functions which are *piecewise* C^1 in the following sense: one period has a finite partition, and on each open piece both f and f' have continuous extensions to the closed piece. Adjacent extensions may have different endpoint values. Values assigned at partition points may be chosen separately, consistently with periodicity. Integrals of such a function are sums of the ordinary integrals of the continuous extensions on those pieces. Changing the finitely many point values leaves these integrals unchanged.

For integrable real functions g and h, *orthogonality on* $[-\pi,\pi]$ means $\int_{-\pi}^{\pi} gh\,\mathrm{d}x = 0$. The functions 1, $\cos nx$ and $\sin nx$, for positive integers n, are mutually orthogonal. More precisely, for positive integers m, n,

$$\int_{-\pi}^{\pi} \cos nx \cos mx\,\mathrm{d}x = \int_{-\pi}^{\pi} \sin nx \sin mx\,\mathrm{d}x = \begin{cases} \pi, & m = n, \\ 0, & m \neq n, \end{cases} \tag{35.2}$$

and every mixed sine–cosine integral is zero. The integrals of sine or cosine alone vanish, while the integral of 1 is 2π.

To prove these facts, for a nonzero integer k the sine and cosine primitives give $\int_{-\pi}^{\pi} \cos kx\,\mathrm{d}x = 0$ and $\int_{-\pi}^{\pi} \sin kx\,\mathrm{d}x = 0$. The identities $2\cos nx\cos mx = \cos((n-m)x) + \cos((n+m)x)$ and $2\sin nx\sin mx = \cos((n-m)x) - \cos((n+m)x)$ give Eq. (35.2). When $m = n$, the zero-frequency cosine is 1 and contributes $2\pi/2 = \pi$. Otherwise both frequencies are nonzero. Every mixed product is odd, so substitution $x = -u$ shows its integral equals its own negative and is zero. This also proves the stated sine integrals.

A *trigonometric polynomial* is a finite sum $a_0/2 + \sum_{n=1}^{N}(a_n \cos nx + b_n \sin nx)$. Multiplying this sum by a basis function and integrating extracts exactly one coefficient by orthogonality. For a piecewise C^1 function f, define its *Fourier coefficients* by

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,\mathrm{d}x \quad (n \geq 0), \qquad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,\mathrm{d}x \quad (n \geq 1). \tag{35.3}$$

Here $\cos 0x = 1$ and the constant term is $a_0/2$. The *Fourier partial sum* is $S_N(x) = a_0/2 + \sum_{n=1}^{N}(a_n \cos nx + b_n \sin nx)$. For each of the basis functions through frequency N, the integral of $(f - S_N)$ times that basis function is zero. Indeed, the coefficient formula gives its integral against f, and orthogonality gives the same integral against S_N. This finite identity requires no convergence or exchange of an infinite sum with an integral.

The same orthogonality gives an approximation property without an infinite convergence theorem. Define the *mean squared error* of a trigonometric polynomial T by

$$E(T) = \frac{1}{2\pi}\int_{-\pi}^{\pi} (f(x) - T(x))^2\,\mathrm{d}x.$$

Among all polynomials using frequencies through N, the Fourier partial sum S_N uniquely minimises this error. To prove the assertion, write another such polynomial as $T = \alpha_0/2 + \sum_{n=1}^{N}(\alpha_n \cos nx + \beta_n \sin nx)$. Expanding the square of $f - T = (f - S_N) + (S_N - T)$ gives

$$E(T) = E(S_N) + \frac{1}{2\pi}\int_{-\pi}^{\pi} (S_N - T)^2\,\mathrm{d}x.$$

The cross term vanishes because $S_N - T$ is a finite linear combination of the basis functions just shown orthogonal to $f - S_N$. Expand the remaining finite square. Every mixed basis integral is zero, and the diagonal integrals have the values in Eq. (35.2). Consequently

$$E(T) = E(S_N) + \frac{(a_0 - \alpha_0)^2}{4} + \frac{1}{2}\sum_{n=1}^{N}\bigl((a_n - \alpha_n)^2 + (b_n - \beta_n)^2\bigr).$$

Each added term is nonnegative. Equality requires every coefficient to agree, and hence $T = S_N$. This proves both minimality and uniqueness. The error measures an average over a period. Its minimisation does not assert a small error at every point, especially near a jump.

A real function is *even* if $f(-x) = f(x)$ and *odd* if $f(-x) = -f(x)$ for every x. For an even function, every b_n is zero and $a_n = (2/\pi)\int_0^{\pi} f(x)\cos nx\,\mathrm{d}x$. For an odd function, every a_n is zero and $b_n = (2/\pi)\int_0^{\pi} f(x)\sin nx\,\mathrm{d}x$. These statements follow from the parity of the products and substitution in the negative half of the integral. For an even integrand that half equals the positive half; for an odd integrand it is its negative.

35.4 The series of a periodic linear ramp

Define a periodic function by $f(x) = x$ for $-\pi < x < \pi$ and $f(\pi) = f(-\pi) = 0$, then extend with period 2π. It is odd, so $a_n = 0$ for every $n \geq 0$. For $n \geq 1$, integration by parts gives

$$b_n = \frac{2}{\pi}\int_0^{\pi} x \sin nx \, dx = \frac{2}{\pi}\left(\left[-\frac{x\cos nx}{n}\right]_0^{\pi} + \frac{1}{n}\int_0^{\pi}\cos nx \, dx\right) = \frac{2(-1)^{n+1}}{n}.$$

In particular $b_1 = 2$ and $b_2 = -1$. The associated series is $2\sum_{n=1}^{\infty}(-1)^{n+1}\sin nx/n$. The coefficient computation alone has not yet identified its sum. For one retained harmonic the best approximation is $S_1(x) = 2\sin x$. Its mean squared error can be evaluated by finite integration:

$$E(S_1) = \frac{1}{2\pi}\int_{-\pi}^{\pi}(x - 2\sin x)^2 \, dx = \frac{\pi^2}{3} - 2.$$

Here the three integrated terms are $2\pi^3/3$, -8π and 4π, respectively, using the coefficient computation and sine orthogonality. The best constant is zero, with error $\pi^2/3$. Thus keeping the first harmonic reduces the mean squared error by exactly 2. This finite approximation statement does not yet identify the infinite series.

Theorem 35.2 (Fourier convergence, cited)**.** *For a real 2π-periodic piecewise C^1 function with the finite-partition hypotheses above, its Fourier partial sums satisfy*

$$\lim_{N\to\infty} S_N(x) = \frac{f(x^-) + f(x^+)}{2} \qquad (x \in \mathbb{R}),$$

where the two values denote the one-sided limits of the periodic function. At a point of continuity the limit is $f(x)$. This is the piecewise smooth Fourier convergence theorem [C].

The cited source is Orloff's MIT ES.1803 Topic 22 notes, §22.5, p. 4. Our finite closed-piece C^1 hypotheses give a sufficient piecewise smooth case. The theorem makes the ramp series equal to x for $-\pi < x < \pi$. At $x = \pi$ the periodic one-sided values are π and $-\pi$, so the sum is zero. Each partial sum also vanishes there because $\sin n\pi = 0$. Arbitrary reassignment of the function at a jump would not alter its coefficients or this midpoint conclusion.

Figure 35.2 compares finite sums with the ramp. The limiting midpoint rule concerns the value at a jump, not a uniform error bound nearby.

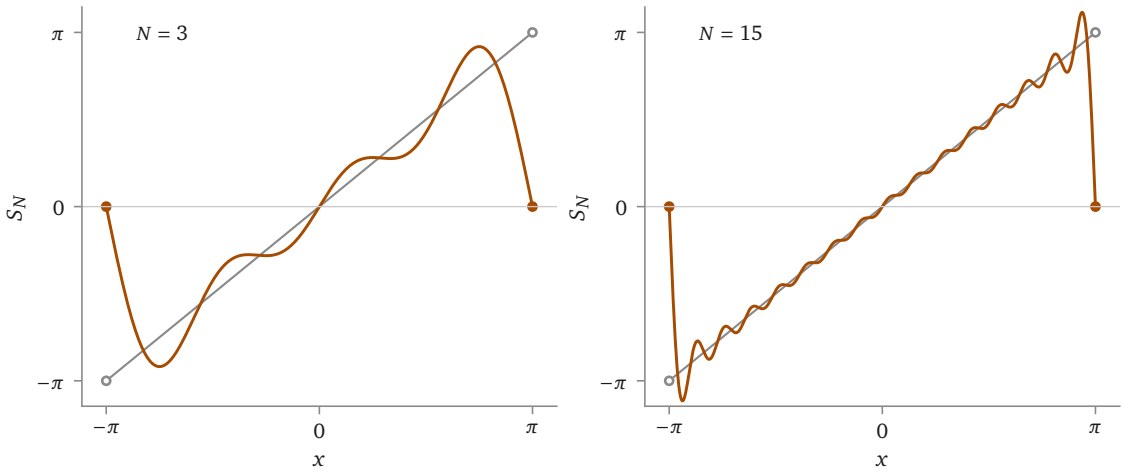


Figure 35.2: Fourier partial sums with $N = 3$ and $N = 15$ for the periodic ramp $-\pi < x < \pi$. At the endpoint jumps all partial sums equal zero.

The convergence cannot be uniform on a full period containing a jump. Each partial sum is continuous, whereas the limiting function with midpoint values at the jumps is discontinuous. Lecture 30's uniform-limit continuity theorem excludes uniform convergence there. The coefficient and convergence statements answer different questions: the former constructs the sums, and the latter identifies their pointwise limit.

35.5 Complex coefficients and symmetric partial sums

For an integer $n \in \mathbb{Z}$, define the complex coefficient

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-\mathrm{i}nx} \, \mathrm{d}x.$$

A complex integral here means the integrals of its real and imaginary parts combined as one complex number. The exponential identity $e^{-\mathrm{i}nx} = \cos nx - \mathrm{i} \sin nx$ gives

$$c_0 = \frac{a_0}{2}, \qquad c_n = \frac{a_n - \mathrm{i}b_n}{2}, \quad c_{-n} = \frac{a_n + \mathrm{i}b_n}{2} \qquad (n \geq 1).$$

For real f, the coefficients satisfy $c_{-n} = \overline{c_n}$. Combining the terms with opposite indices gives the exact finite identity

$$S_N(x) = \sum_{n=-N}^{N} c_n e^{\mathrm{i}nx}.$$

Thus the notation $\sum_{n\in\mathbb{Z}} c_n e^{\mathrm{i}nx}$ denotes the limit of these symmetric partial sums. It does not prescribe two separately summed infinite tails. For the ramp, $c_0 = 0$ and $c_n = \mathrm{i}(-1)^n/n$ for every nonzero integer n, as follows by substituting its real coefficients.

35.6 Finite Fourier forcing

Return to $y'' + 2\gamma y' + \omega_0^2 y = f(t)$ with $\gamma > 0$ and $\omega_0 > 0$. Suppose the real forcing is a trigonometric polynomial, written as $f(t) = \sum_{n=-M}^{M} d_n e^{\mathrm{i}nt}$, where $M \geq 0$ is an integer and

$d_{-n} = \overline{d_n}$. Each $P(in)$ is nonzero: the zero mode gives ω_0^2, and each nonzero mode has imaginary part $2\gamma n \neq 0$. Finite differentiation and addition of the exponential responses give

$$y_{\mathrm{per}}(t) = \sum_{n=-M}^{M} \frac{d_n}{P(in)} e^{int}. \tag{35.4}$$

Applying the differential expression to each term yields $d_n e^{int}$, so the sum solves the required equation. The negative-index terms are conjugates of the positive-index terms, because $P(-in) = \overline{P(in)}$. The zero coefficient is real. Hence y_{per} is a real 2π-periodic solution.

Every other solution differs by a homogeneous motion tending to zero. If another solution were also 2π-periodic, their difference h would be periodic and tend to zero. For any fixed t, $h(t) = h(t + 2\pi j) \to 0$ as the integer $j \to \infty$. Thus $h(t) = 0$ for every t, proving uniqueness of the periodic response. This argument uses finite forcing; differentiating an arbitrary infinite Fourier series would require additional justification.

For example, with $\gamma = 1$, $\omega_0 = 3$ and $f(t) = 1 + 3\cos 2t - 2\sin t$, the response is

$$y_{\mathrm{per}}(t) = \frac{1}{9} + \frac{15\cos 2t + 12\sin 2t}{41} + \frac{\cos t - 4\sin t}{17}.$$

The forcing coefficients are $d_0 = 1$, $d_1 = \mathrm{i}$, $d_{-1} = -\mathrm{i}$ and $d_2 = d_{-2} = 3/2$. Here $P(\mathrm{i}) = 8 + 2\mathrm{i}$ and $P(2\mathrm{i}) = 5 + 4\mathrm{i}$. The positive-frequency response coefficients in Eq. (35.4) are

$$\frac{d_1}{P(\mathrm{i})} = \frac{1 + 4\mathrm{i}}{34}, \qquad \frac{d_2}{P(2\mathrm{i})} = \frac{15 - 12\mathrm{i}}{82}.$$

Pairing each with its conjugate gives the frequency 1 and frequency 2 terms displayed above; the zero mode gives $1/9$. Their sum is the unique periodic motion, and any initial-data correction decays.

Summary 35.

- For $\gamma > 0$, $\omega_0 > 0$, $F > 0$ and $\omega \geq 0$, the steady sinusoidal response has amplitude $F/\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}$ and the quadrant-aware phase lag defined above.
- Its positive-frequency amplitude peak exists exactly when $0 < \gamma < \omega_0/\sqrt{2}$, at $\sqrt{\omega_0^2 - 2\gamma^2}$. Undamped natural-frequency forcing instead gives growth proportional to $t \sin \omega_0 t$ from rest.
- Orthogonality proves the real coefficient extraction formulas Eq. (35.3). The Fourier series converges to the periodic one-sided midpoint under the finite piecewise C^1 hypotheses of Theorem 35.2 [C].
- Complex Fourier notation uses symmetric partial sums. For real trigonometric-polynomial forcing and positive damping, Eq. (35.4) gives the unique real periodic response.

Sheet 35 treats further signals and oscillator applications, including the quadratic series and its numerical sum. Lecture 36 begins differentiation of functions with several independent real variables.

References

J. Orloff, *ES.1803 Differential Equations*, MIT OpenCourseWare, Spring 2024, Topic 22, §22.5, p. 4, Fourier convergence theorem [C], course notes.